Management of Critically Ill Child

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Age range

Neonate to 18 yrs

500g to 100kg

Objectives

Understand the physiological, anatomical and pathological differences which influence evaluation and treatment

Basic principles of paediatric resuscitation

Important emergency conditions and management principles

General Examination

Skin perfusion – color, warmth, capillary refill (normal < 2 seconds)

Hydration state (depressed anterior fontanelle is a sign of severe dehydration)

Responsiveness – irritability, alertness, response to stimuli, depressed conscious state

Respiratory – tachypnea, bradypnea, grunting, nasal flaring, obligate nose breathing till 6 months of age

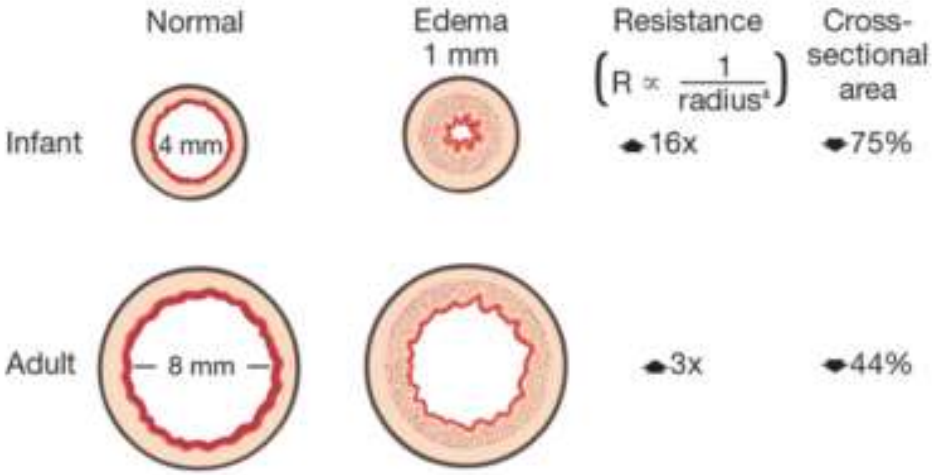
General Examination

Seizures may not always be tonic / clonic may, present as impaired alertness, abnormal vital signs eg fixed & unreactive pupils, abnormal movement eg. cycling or chewing activity

Apart from persistent high fever, Hypothermia can occur in severe sepsis especially in neonate and small child

Respiratory Principles

Biphasic response to hypoxia in neonates, increase respiratory distress first and can easily go into apnoea afterwards



Chest stability/strength – cartilaginous thorax, more horizontal ribs -> diaphragmatic breathing important

Airways – high airway resistance especially during upper airway obstruction -> more common in small children

Less respiratory reserve, higher need for oxygen supplement

Cardiovascular Principles

Small absolute blood volume

Cardiac output dependent on rate due to low stroke volume

Bradycardia is ominous

CVP less accurate, response to fluids bolus similar to adult after 8 wks; Reactive pulmonary vasculature in neonate, more prone to develop pulmonary hypertension (Persistent Pulmonary Hypertension of Newborn, PPHN)

Variable catecholamine response, need titration

Non-hypovolemic Shock

Possible etiologies:

- Pneumothorax
- Pericardial effusion, myocardial dysfunction
- Intestinal ischemia, sepsis
- Adrenal insufficiency
- Pulmonary arterial hypertension
- Congenital heart disease eg coarctation of aorta

Metabolic / Thermal Principles

Greater insensible water loss

Greater heat loss

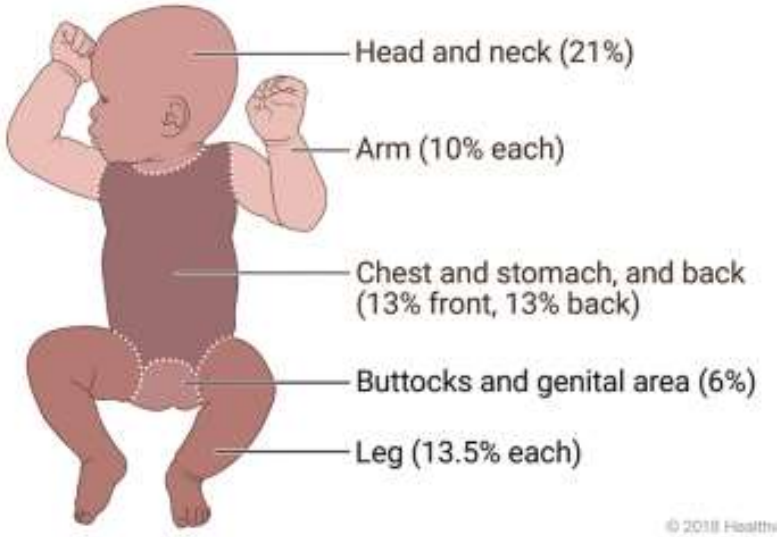
Less reserve, hypoglycemia more common, need higher glucose infusion

Appropriate urine output for age

(neonate 2ml/kg/hr, child 1 ml/kg/hr)

Hypo- and hypernatremia

Hypocalcemia in newborns



Paediatric Early Warning Score (PEWS)

Table 10-1. McCarthy score

OBSERVATION ITEM	SCORE 1 Normal	3 Moderate Impairment	5 Severe Impairment
Quality of cry			
Reaction to pa stimulation			
State variation			
Color			
Hydration			
Response (talk to social overt			

Score = sum of al
<10 only 2.7
>10 but <16
>16 only 92

	0	1	2	3	Score
Behaviour	Playing/ appropriate	Sleeping	Irritable	Lethargic/confused Reduced response to pain	
Cardiovascular	Pink or capillary refill 1-2 seconds	Pale or capillary refill 3 seconds	Grey or capillary refill 4 seconds. Tachycardia of 20 above normal rate	Grey and mottled or capillary refill 5 seconds or above. Tachycardia of 30 above normal rate or bradycardia.	
Respiratory	Within normal parameters, no recession or tracheal tug	>10 above normal parameters, using accessory muscles, 30+% FiO2 or 4+ litres/min	>20 above normal parameters recessing, tracheal tug. 40+% FiO2 or 6+ litres/min	5 below normal parameters with sternal recession, tracheal tug or grunting. 50% FiO2 or 8+ litres/min	
Score 2 extra for 1/4 hourly nebulisers or persistent vomiting following surgery					

Score reference (scoring parameters and action needed may varies between institutions)

0-4 (stable) continue 4hrly assessment

5-6 more frequent assessment, doctor notification needed

> 7 Critical frequent assessment q30min, immediate doctor notification

Summary

Variations in maturation of organ systems and physiologic responses

Child is NOT miniature adult

Physiological condition similar to adult by 8 to 12 years, but not mentally

Body size and surface area constraints

Vascular responses and volume considerations

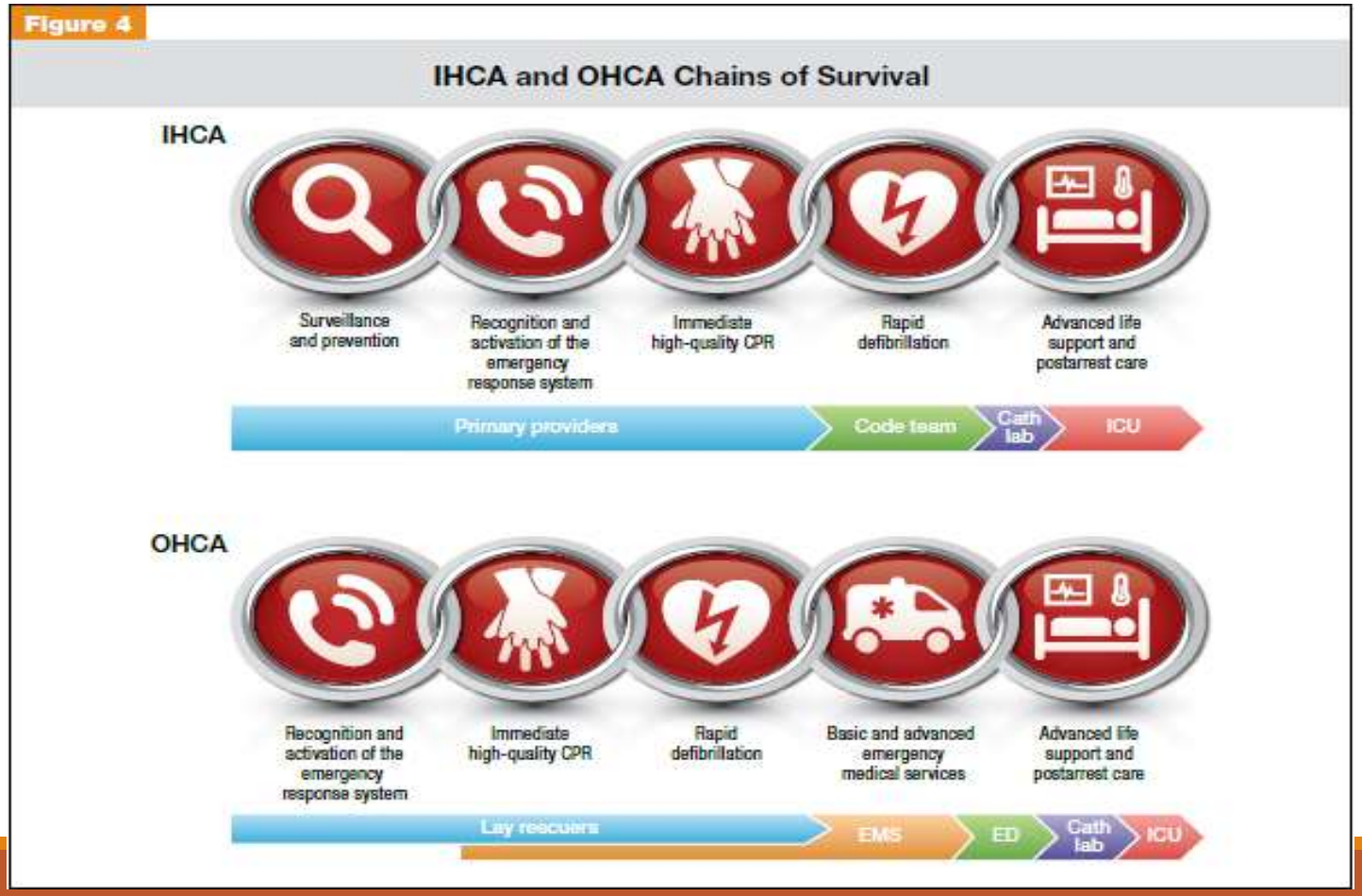
Small "margin for error"

Early consultation

PAEDIATRIC EMERGENCY

Sudden unexpected cessation of functional ventilation and circulation in a person otherwise not expected to die

Respiratory arrest more common than Cardiac arrest



IHCA: In-Hospital Cardiac Arrest

OHCA: Out-Hospital Cardiac Arrest

2022 update on Chain of survival

In addition to Post-Cardiac Arrest Care, highlighted Recovery support

Figure 1. Pediatric Chains of Survival for in-hospital (top) and out-of-hospital (bottom) cardiac arrest.
CPR indicates cardiopulmonary resuscitation.

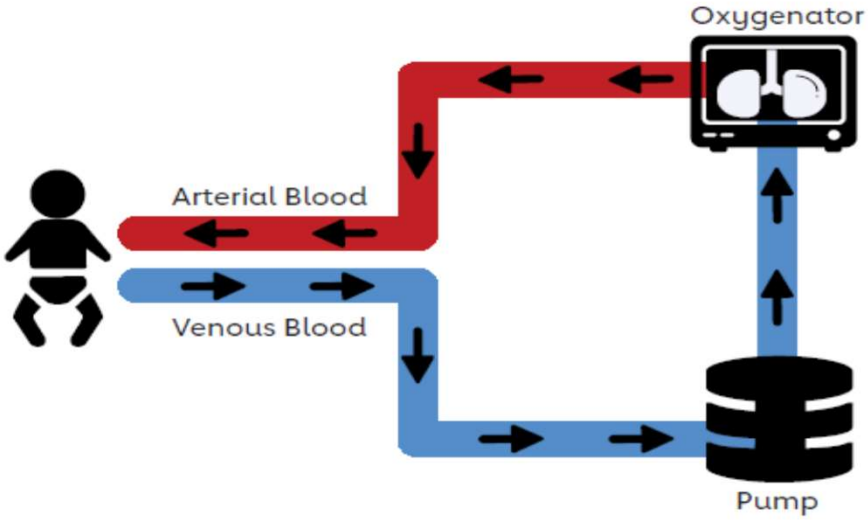
Option of eCPR (Extracorperal Cardiac Pulmonary Resuscitation)

- Improve chance of respond to cardiac arrest
- Long term prognosis still need more reviews, increase risk of CNS complications
- Should consider for readily reversible cases eg witness cardiac arrest, post-op cardiac

Minor updates for Paediatric in 2020

Extracorporeal CPR

Extracorporeal CPR refers to a cardiopulmonary bypass, which **maintains organ perfusion** while cardiac arrest causes are addressed.



2019 Recommendation: Consider extracorporeal CPR in pediatric patients with cardiac diagnoses who are experiencing in-hospital cardiac arrest **in a center with extracorporeal membrane oxygenation capability.**

Because of insufficient evidence, there are **no recommendations for or against extracorporeal CPR** in pediatric **out-of-hospital cardiac arrests** or **noncardiac diagnoses.**

CARDIAC PULMONARY RESUSCITATION (CPR)

Phase 1 - BASIC CPR

- aim at maintaining Oxygenation and Circulation
  - Airway
  - Breathing
  - Circulation
- then

Phase 2 - ADVANCE LIFE SUPPORT

- aim at restoration of Spontaneous Circulation
  - Drugs
  - ECG
  - Fibrillation
- then

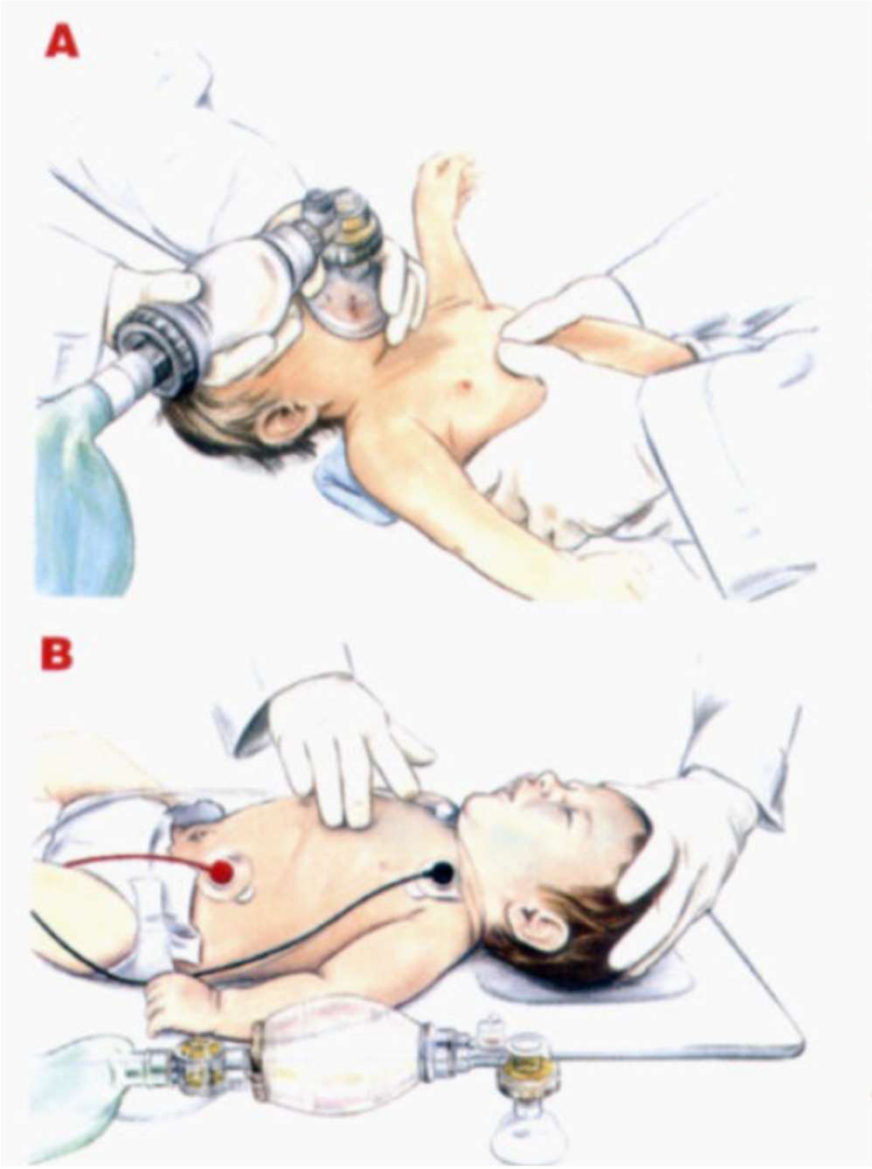
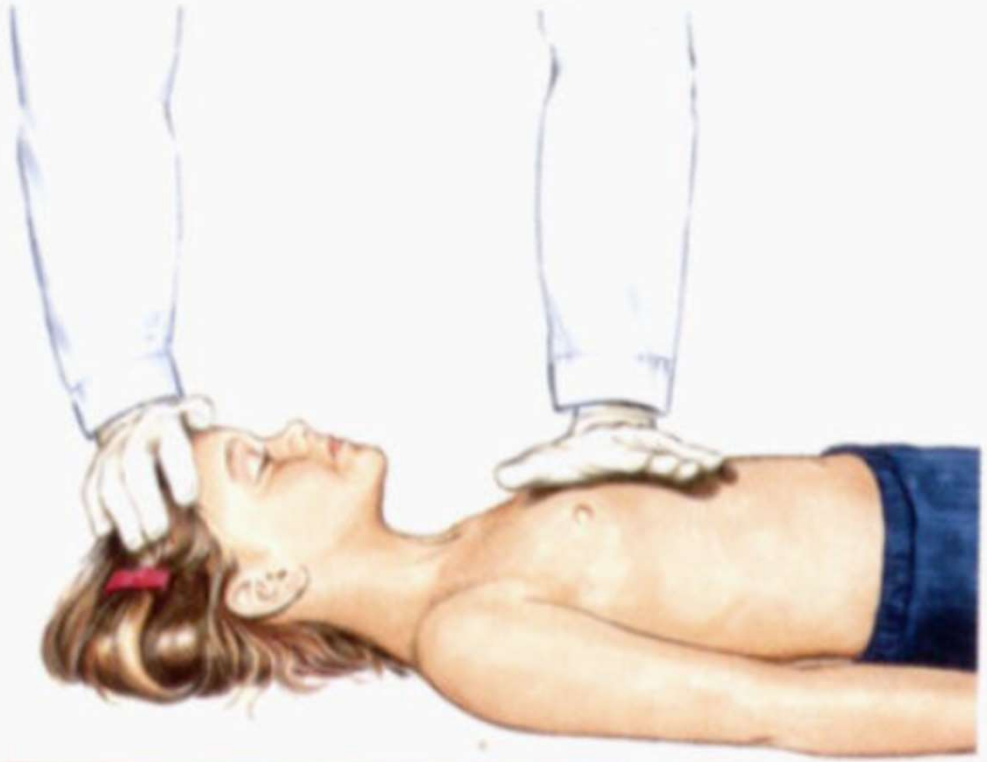


FIGURE 18. One-hand chest compression technique for child.



From AHA PALS Provider Manual

Phase 3 - PROLONGED LIFE SUPPORT

- management of multiple organ failure
- especially on the aspect of BRAIN RESUSCITATION
- Gauging - cause of arrest and assess condition
- Human Mentation - Cerebral Resuscitation
- Intensive Care Support

Priority should be given to the establishment of A.B.C.

Good team-work essential

Good communication and close loop practice

Good documentation of the events

Concept change in CPR

Airway Breathing Circulation

Circulation Airway Breathing

Time Difference – first ventilation after 30 compressions (delay of around 20 seconds)

Emphasis on Circulation is MOST important

RATE, Depth of compression, Minimum interruption

Better choice for lay rescuer as opening Airway and rescue Breath is a more difficult procedure

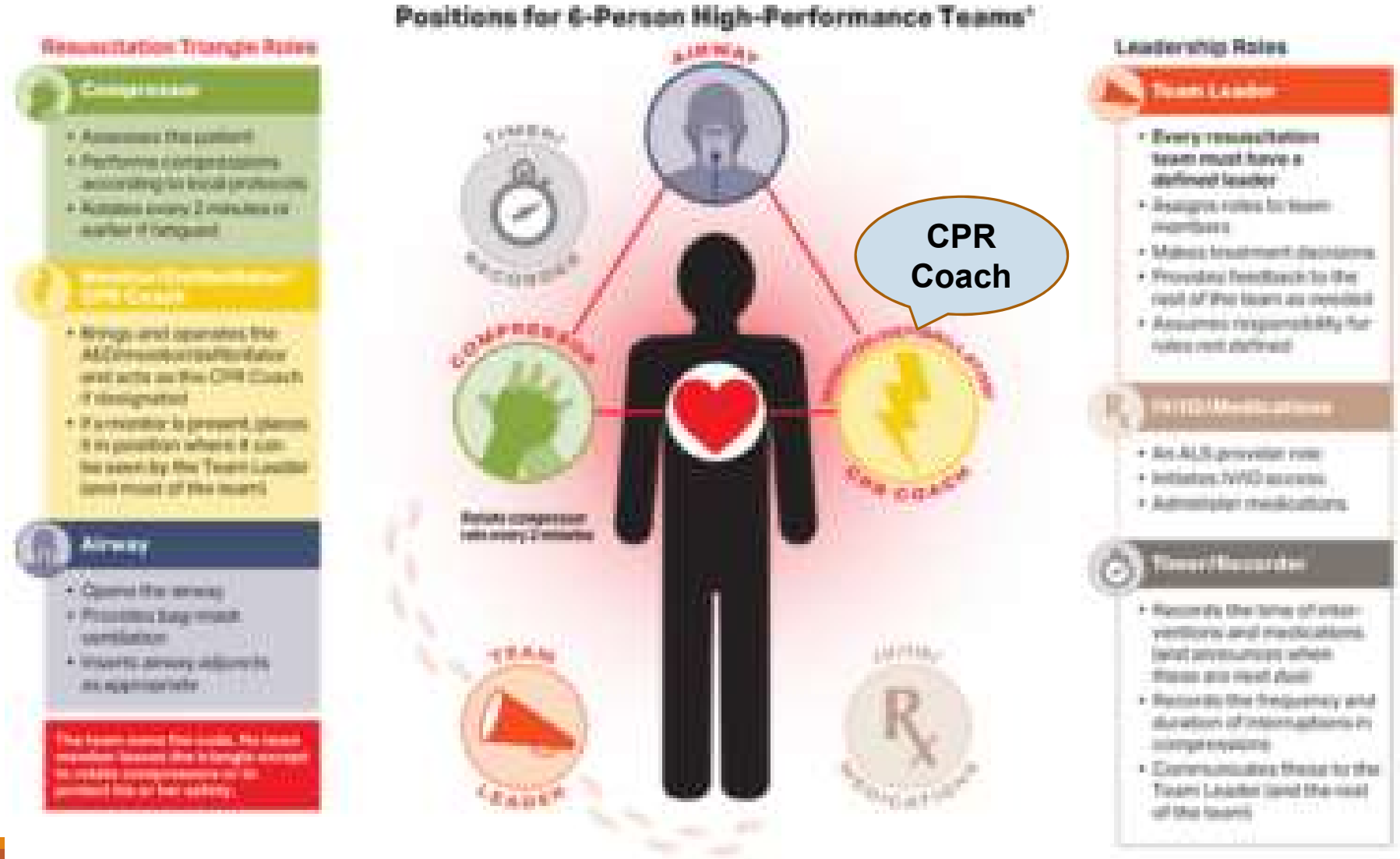
Simplify BLS by recommend Compression Only for untrained lay rescuer. Option for trained rescuer also

UK guidelines still support rescue breath

Gift of life: organ donation should be considered in those who do not have ROSC, brain death or withdrawal of care

2024 recommend Respiratory Rate in infant less than 1 year should be at least 30/min and 25/min for 1 year or older.

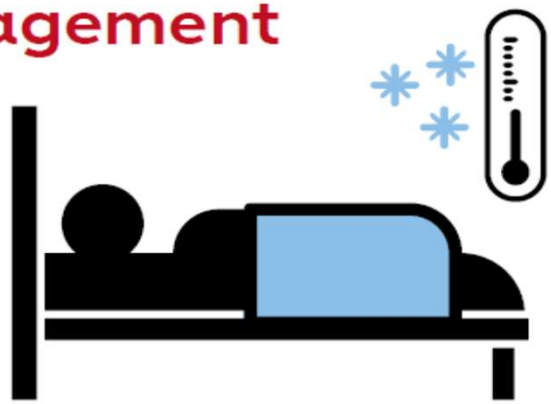
Team Work


Positions for 6-Person High-Performance Teams*
Resuscitation Triangle Roles
Airway
Leadership Roles
Team Leader
CPR Coach

Minor updates for Paediatric in 2020

Targeted Temperature Management

Targeted temperature management (TTM) involves **keeping core temperatures within a certain range** to induce **therapeutic hypothermia** in pediatric patients who remain comatose after cardiac arrest.

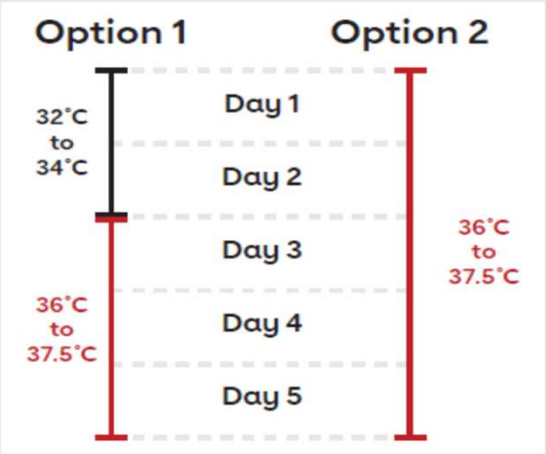


2019 Recommendation: Continuously monitor core temperature during TTM.

For patients between 24 hours and 18 years of age who remain comatose after cardiac arrest, **there are 2 reasonable options:**

Option 1: Use TTM to maintain 32°C to 34°C, followed by TTM to maintain 36°C to 37.5°C.

Option 2: Use TTM to maintain 36°C to 37.5°C.



2019 Recommendation: TTM can be considered for both in-hospital and out-of-hospital cardiac arrest.

New TTM recommendations are based on the **THAPCA-IH trial** (Therapeutic Hypothermia After Pediatric Cardiac Arrest In-Hospital).

Table 2 Summary of High-Quality CPR Components for BLS Providers

Component	Adults and Adolescents	Children (Age 1 Year to Puberty)	Infants (Age Less Than 1 Year, Excluding Newborns)
Scene safety	Make sure the environment is safe for rescuers and victim		
Recognition of cardiac arrest	Check for responsiveness No breathing or only gasping (ie, no normal breathing) No definite pulse felt within 10 seconds (Breathing and pulse check can be performed simultaneously in less than 10 seconds)		
Activation of emergency response system	If you are alone with no mobile phone, leave the victim to activate the emergency response system and get the AED before beginning CPR Otherwise, send someone and begin CPR immediately; use the AED as soon as it is available	***Witnessed collapse*** Follow steps for adults and adolescents on the left ***Unwitnessed collapse*** Give 2 minutes of CPR Leave the victim to activate the emergency response system and get the AED Return to the child or infant and resume CPR; use the AED as soon as it is available	
Compression-ventilation ratio *without advanced airway*	***1 or 2 rescuers*** 30:2	***1 rescuer*** 30:2 ***2 or more rescuers*** 15:2	
Compression-ventilation ratio *with advanced airway*	Continuous compressions at a rate of 100-120/min Give 1 breath every 6 seconds (10 breaths/min)		
Compression rate	100-120/min		
Compression depth	At least 2 inches (5 cm)*	At least one third AP diameter of chest About 2 inches (5 cm)	At least one third AP diameter of chest About 1½ inches (4 cm)
Hand placement	2 hands on the lower half of the breastbone (sternum)	2 hands or 1 hand (optional for very small child) on the lower half of the breastbone (sternum)	***1 rescuer*** 2 fingers in the center of the chest, just below the nipple line ***2 or more rescuers*** 2 thumb–encircling hands in the center of the chest, just below the nipple line
Chest recoil	Allow full recoil of chest after each compression; do not lean on the chest after each compression		
Minimizing interruptions	Limit interruptions in chest compressions to less than 10 seconds		

*Compression depth should be no more than 2.4 inches (6 cm).

Abbreviations: AED, automated external defibrillator; AP, anteroposterior; CPR, cardiopulmonary resuscitation.

Neonatal Resuscitation

Golden Minute (60-second) mark for completing the initial steps, reevaluating, and beginning ventilation (if required)

Apart from SpO2 monitor, additional ECG monitoring is recommended

Suctioning non-vigorous infants with MSL approach not recommended.

Avoiding unnecessary delay in initiation of ventilation

Delayed cord clamping for longer than 30 seconds is reasonable for both term and preterm infants who do not require resuscitation at birth

Targeted pre-ductal SpO_2 after birth

1 min	60-70%
2 min	65-85%
3 min	70-90%
4 min	75-90%
5 min	80-90%
10 min	85-90%

FLUID ADMINISTRATION DURING RESUSCITATION

Intra-osseous route

- For volume bolus and effective delivery of medication
- method : same as bone marrow puncture
- disposable needle available
- site: usually ant. tibial because more convenient

Others

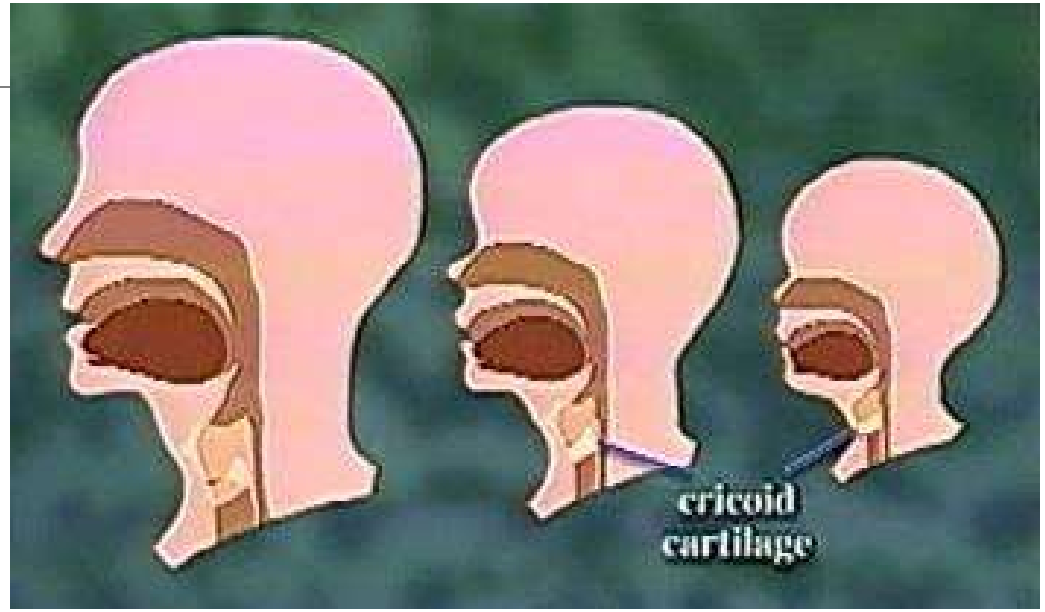
Cuffed endotracheal tubes can be used in infants and children provided correct tube size and cuff inflation pressure are used

Exhaled CO2 detection is recommended for confirmation of endotracheal tube placement

Problems with Paediatric Intubation

Anatomical difference

- Airway smaller and shorter than adult
- Adult larynx more cylindrical with narrowest at vocal cord



  - Child: funnel shape, narrowest at cricoid cartilage
  - More superior and anterior Tongue and epiglottis relatively large

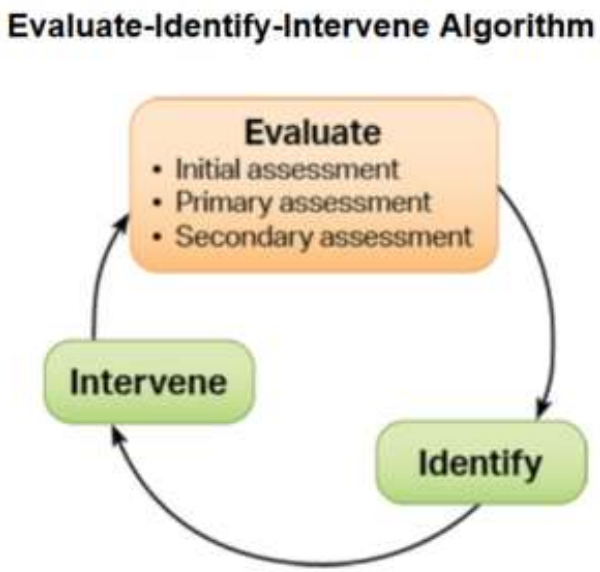
NON-Cuffed Endotracheal tube commonly used in the past but with better designed cuffed endotracheal tube, getting popular

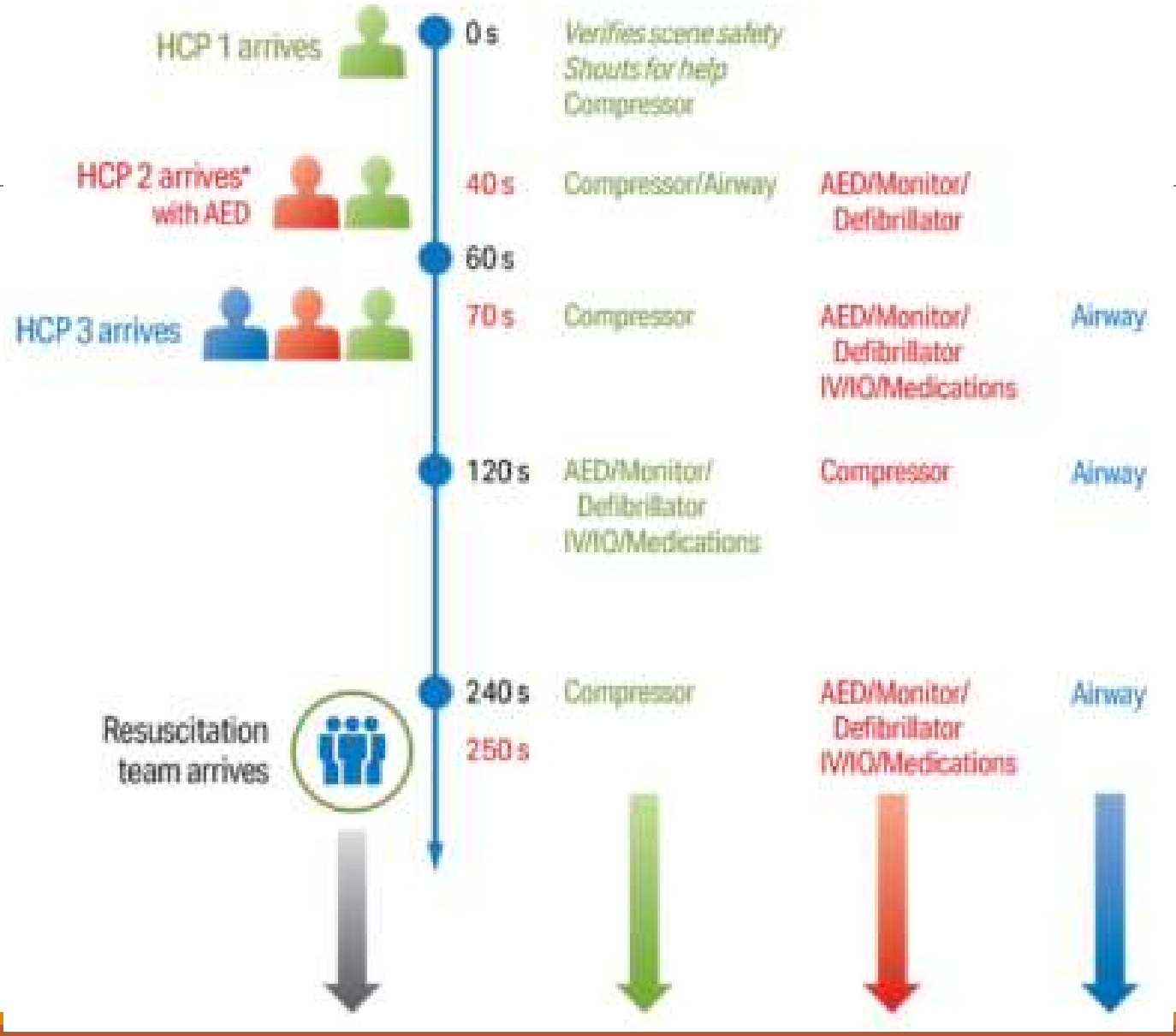
Initial Assessment – Paediatric Assessment Triangle (can be completed in less than seconds)

Appearance: Abnormal tome, Decrease interaction, Inconsolable, Abnormal look / gaze, Abnormal speech / cry

Work of breathing: Abnormal sounds, Abnormal positioning, Retractions, Flaring, Apnoea / gasping

Circulation to Skin: Pallor, Mottling, Dusky, Cyanosis




HCP 1 arrives
0 s
Verifies scene safety
Shouts for help
Compressor
HCP 2 arrives*
with AED
40 s
Compressor/Airway
AED/Monitor/
Defibrillator
60 s
HCP 3 arrives
70 s
Compressor
AED/Monitor/
Defibrillator
IV/IO/Medications
Airway
120 s
AED/Monitor/
Defibrillator
IV/IO/Medications
Compressor
Airway
240 s
250 s
Compressor
AED/Monitor/
Defibrillator
IV/IO/Medications
Airway
Resuscitation
team arrives

Laryngeal Mask

More and more studies in children showed More studies in children showeduring cardiac arrest in the past

Higher complication rate in smaller children due to ? experience

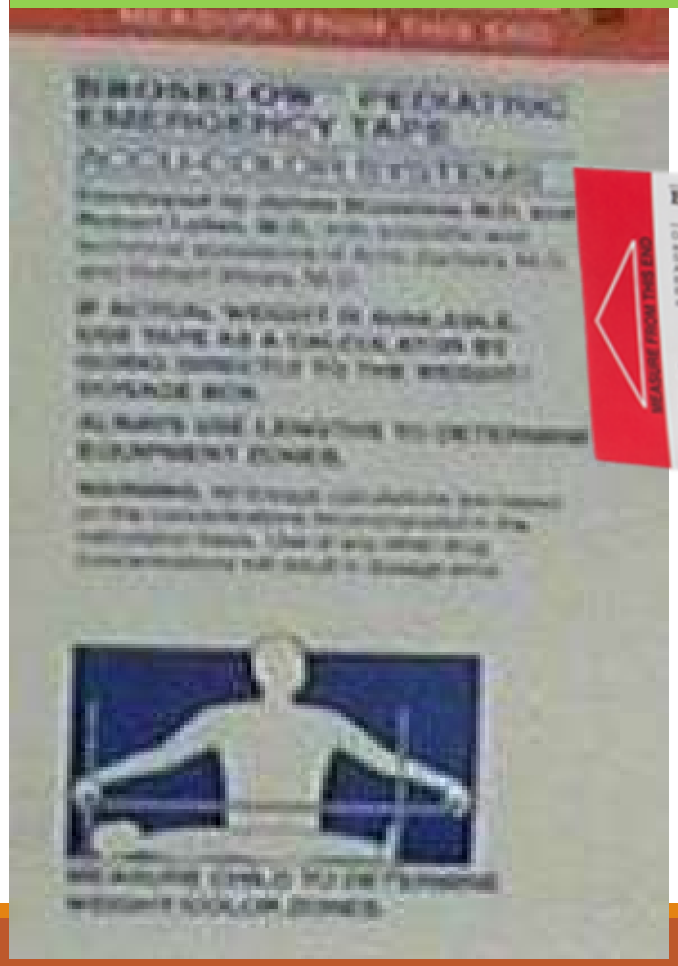
Can be helpful for management of difficult airway

Treatment Recommendation

- acceptable initial alternative airway adjunct for providers during pediatric cardiac arrest when tracheal intubation is difficult to achieve

Quick References

Broselow Tape

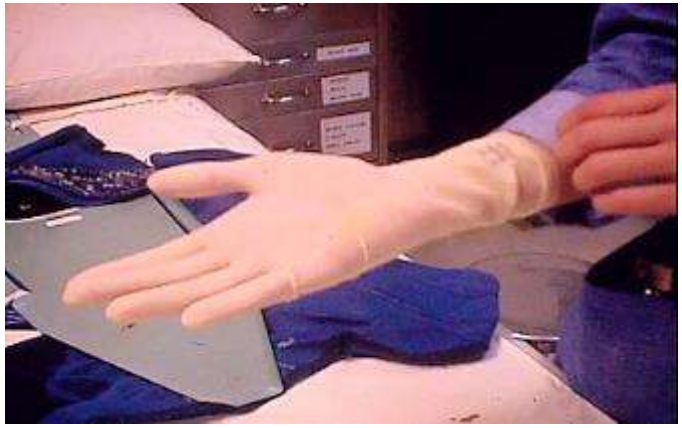


Pediatric Resuscitation Quick Reference

Equipment	Premature Neonate (< 1kg)	Newborn/ Small Infant (3-5 kg)	1 year (6-9 kg)	Toddler (10-11 kg)	Small Child (12-14 kg)	Child (15-18 kg)	Child (19-22 kg)	Large Child (24-30 kg)	Adult (≥32 kg)
Resuscitation bag	Infant		Child	Child	Child		Child	Child/adult	Adult
O_2 mask	Newborn		Newborn	Pediatric	Pediatric		Pediatric	Adult	Adult
Oral airway	Infant/small child		Infant/small child	Small child	Child		Child/small adult		Medium adult
Laryngoscope blade (size)	0 straight	0-1 straight	1 straight		2 straight	2 straight or curved	2 straight or curved	2-3 straight or curved	3 straight or curved
Tracheal tube (mm)	2.5 uncuffed	3.0-3.5 uncuffed	3.5 uncuffed	4.0 uncuffed	4.5 uncuffed	5.0 uncuffed	5.5 uncuffed	6.0 cuffed	6.5 cuffed
Tracheal tube length (cm at lip)	6 - 8	10-10.5	10-10.5	11-12	12.5-13.5	14-15	15.5-16.5	17-18	18.5-19.5
Stylet	Small (2.2mm)		Small (2.2mm)		Small (2.2mm)	Medium (4mm)	Medium (4mm)	Large (5mm)	Large (5mm)
Laryngeal Mask (volume)		Size 1 (4ml)	Size 1.5 (7ml)		Size 2 (10ml)		Size 2.5 (15ml)		Size 4 (30ml)
Suction catheter (F) 2 x TT size	6-8		8	8-10	10		10		12
Nasogastric tube (F)	5-8		5-8	8-10	10	10-12	12-14	14-18	18
Urinary catheter (F)	5-8		5-8	8-10	10	10-12	10-12	12	12
Chest tube (F)	10-12		10-12	16-20	20-24		24-32	28-32	32-40
Adenosine 0.1mg/kg -> 0.2mg/kg		0.3mg -> 0.6mg	1mg -> 2mg		1.5 mg -> 3 mg		2mg -> 4mg	3mg -> 6mg	6mg ->12mg
Adrenaline (1:10,000) 0.1ml/kg ET: 3X IV dose	0.1ml	0.3 ml	1 ml		1.5 ml		2 ml	3 ml	5 ml
Amiodarone (5mg/kg, max 3 dose or 300mg)		20mg	50mg		75mg		100mg	135mg	300mg
Atropine		0.1mg	0.2mg		0.3mg		0.4mg	0.6mg	1mg
10% Calcium Gluconate 0.5ml/kg		2ml	5ml		7.5ml		10ml	12ml	20ml
Insulin Dextrose 25u in 500ml D20 (0.1u/2ml)		8ml	10ml		15ml		20ml	30ml	50ml
Lignocaine 1% (1mg/kg)		4mg	5mg		7.5mg		10mg	12mg	50mg
Sodium bicarbonate 8.4% 1ml/kg	1 ml	3 ml	10 ml		15 ml		20 ml	27 ml	50 ml
Midazolam (0.1mg/kg)	0.1 mg	0.4 mg	1mg		1.5mg		2mg	2.7mg	5mg
Lorazepam (0.05mg/kg)		0.2mg	1mg		1.5mg		2mg	2.7mg	4mg
Rocuronium (0.6mg/kg)	0.6 mg	2.5 mg	6mg		12mg		12mg	15mg	30mg
Flumazenil			0.1mg		0.15mg		0.2mg	0.2mg	0.2mg
Naloxone		0.4mg	1mg		1.5mg		2mg	2mg	2mg
Paraldehyde (rectal)		1.2ml	3ml		5ml		10ml		10ml
Valium (rectal)		2.5mg	3.5mg		7.5mg		10mg		10mg
Defibrillation / cardioversion external paddles	Infant paddles		Adult paddles when ≥1 yr or ≥10kg		Adult paddles		Adult paddles		Adult paddles
Synchronised cardioversion (1 joule/kg)		3 J	7 J	10 J	20 J		30 J		50 J
Defibrillation (4 J/kg)		20 J	30 J		50 J		100 J		200 J

SARS / COVID-19

Personal Protective Equipment



Universal Precaution

COVID-19 Pandemic

Different ball game, shooting target

Initial outbreak – prognosis better than SARS but widespread

Delta variant, and Omicron related issues

Patient Isolation

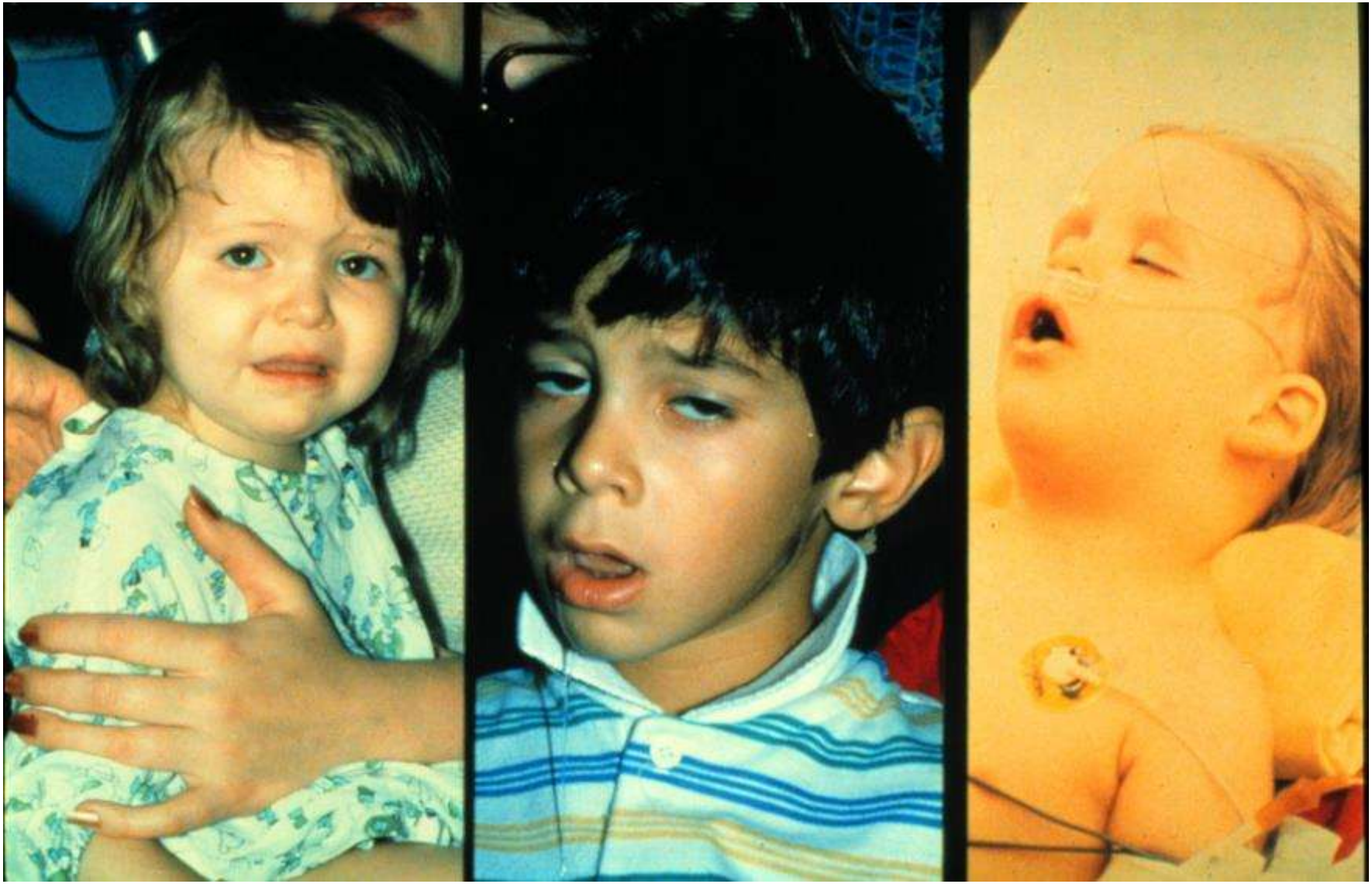
1. Nurse in Airborne Infection Isolation Room (AIIR) (i.e. with negative pressure and at least 12 ACH) en-suite with toilet facility, in an isolation ward setting)

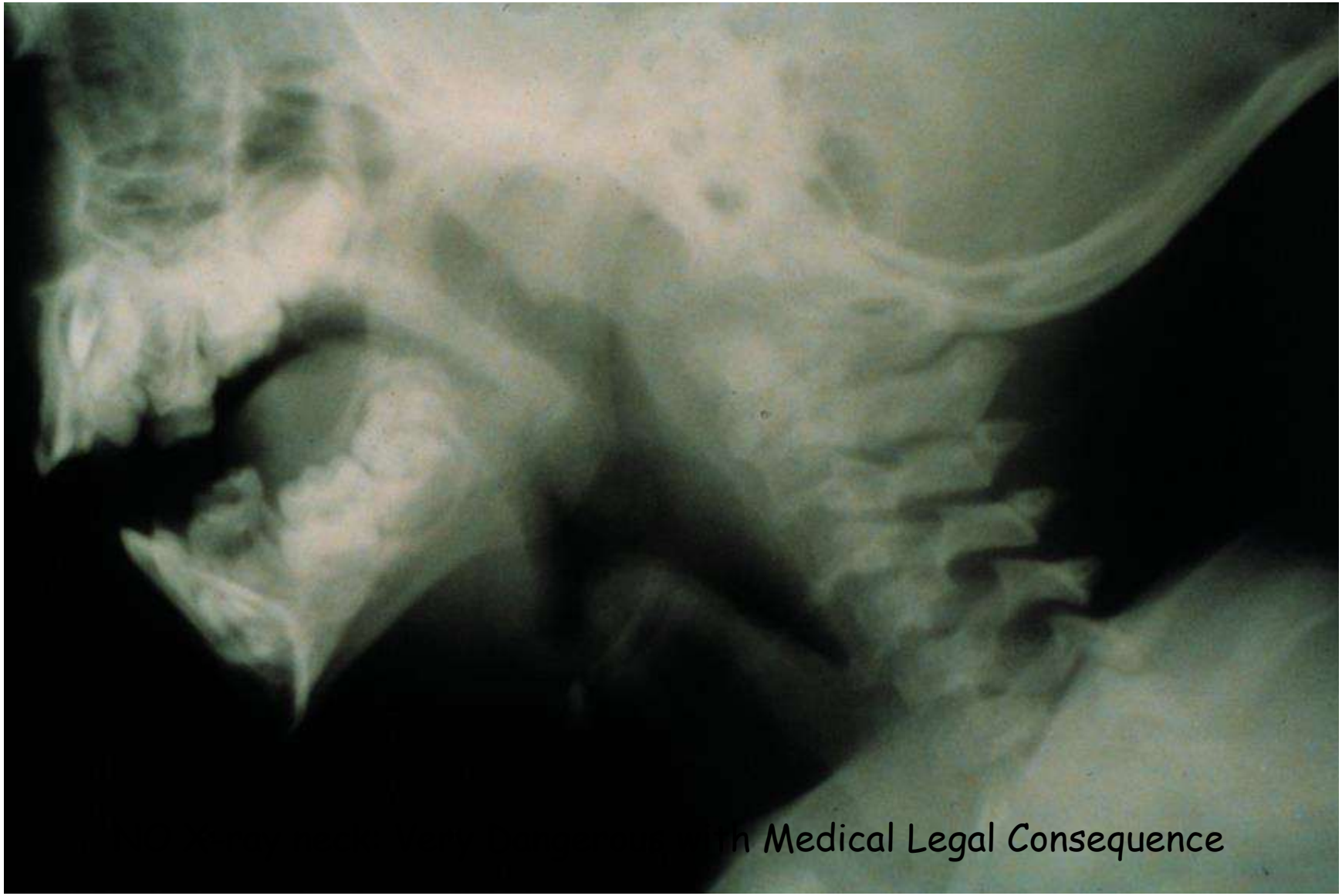
2. Implement Airborne, Droplet and Contact Precautions in addition to Standard Precautions

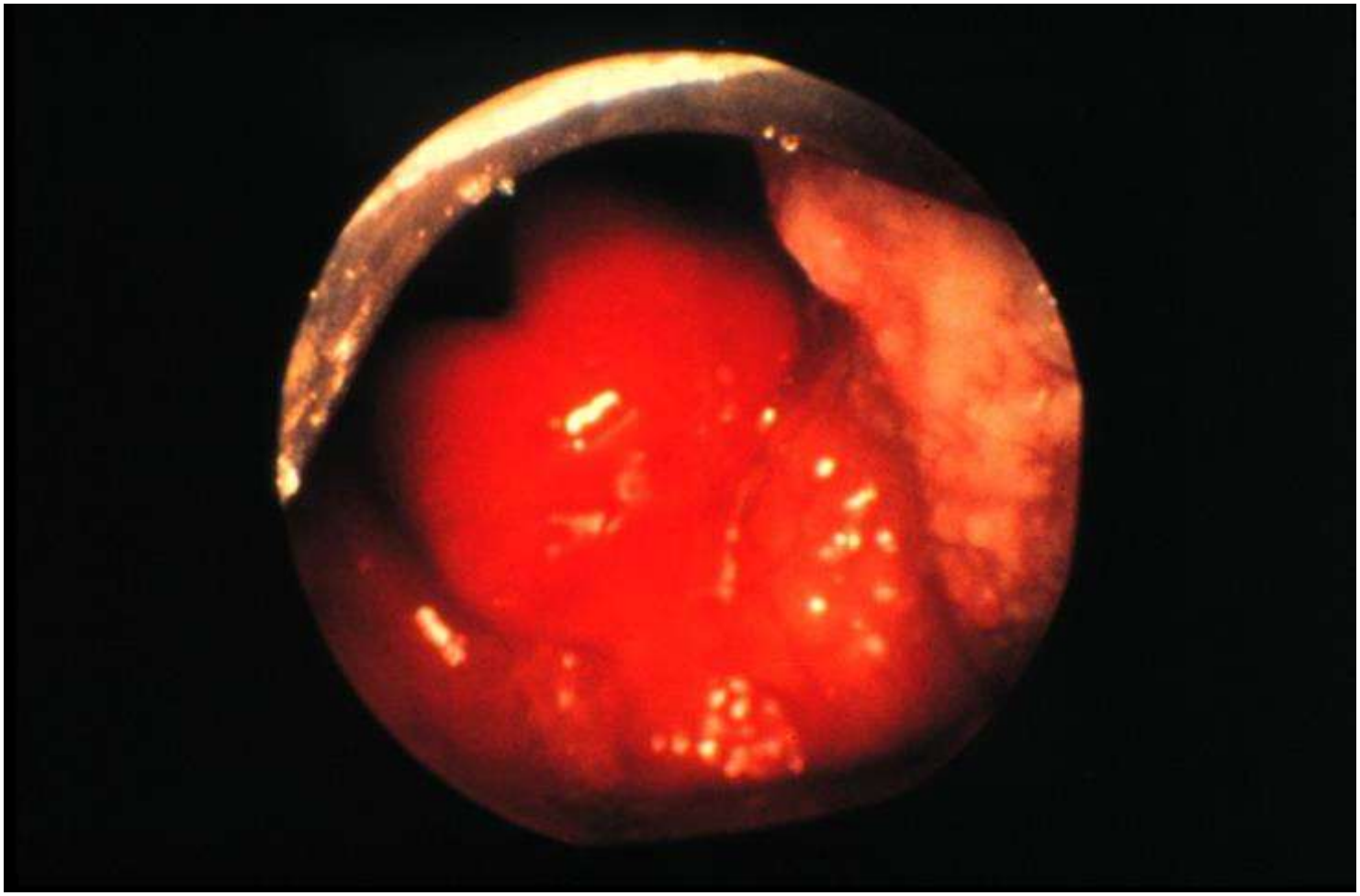
3. PPE: N95 respirator, face shield / goggles / eye visors, gown, gloves, and cap (optional) for aerosol-generating procedures and routine patient care

ACUTE EPIGLOTTIS AND CROUP

	CROUP	ACUTE EPIGLOTTIS
Age	6 mths to 3 yrs	2-6 yrs
Onset	gradual	acute
Etiology	viral	H. influenza B
Swelling	subglottic	supraglottic
Symptoms		
Cough voice	hoarse cough	muffled voice
Fever	absent to high	high
Appearance	not acutely ill	anxious, toxic
Larynx	non-tender	tender
Recurrence	may recur	rare
Seasonal	winter	none




n Medical Legal Consequence



MANAGEMENT OF ACUTE EPIGLOTTIS

All suspected cases - emergency

- monitor vital sign
- oxygen supplement even if patient is not cyanotic
- NO throat examination
- do not place patient in horizontal position
- X-ray neck is DANGEROUS and is not necessary for confirmation of diagnosis
- transport patient in sitting with experience doctor ready for intubation.
- Airway equipment
  - bag and mask
  - laryngoscope
  - ET tubes just 1 size smaller than recommended
  - percutaneous tracheostomy set

MANAGEMENT OF ACUTE EPIGLOTTIS

Supportive

- fluid and hydration
- treatment of post obstructive pulmonary oedema
- sedation and avoid accidental extubation
- care of ET tube
- adequate humidification

Extubation

- general condition
- fever subsiding
- presence of air leak
- usually done at 18-24 hours

STATUS EPILEPTICUS (SE)

- 3 seizures without awakening or continuous motor seizure activity for more than 30 minutes
- Treatment should be given for prolonged seizures of > 5 minutes

General principle of treatment

Prolonged Seizure

- Duration > 5 minutes
- Spontaneous cessation unlikely if duration > 5 – 10 minutes
- Prolonged seizure more likely progress to status
- Start treatment (prehospital / AED)

Initial / Early Convulsive Status Epilepticus

- Duration 20-30 minutes
- Emergency management and seizure control important
- Better response to first medications
  - > 80% within 30min
  - 75% within 60min
  - 65% within 90min

General principle of treatment

Established Convulsive SE

- Duration 30-60 minutes
- Additional anti-epileptic drugs
- More close monitoring, investigations for underlying causes and complications

Refractory SE

- Unresponsive to 2 anti-epileptic drugs with duration > 60 minutes
- Associated with high morbidity and mortality
- Required ICU admission and aggressive control, ventilatory / haemodynamic support

Resuscitation and stabilization

Basic life support

Airway, Breathing & Circulation

Ensure adequate oxygenation

Intubation is not usually necessary but be prepared

Pre-hospital treatment

General believe earlier treatment is more likely to stop the seizure than those late treatment

Most support single pre-hospital dose of rectal diazepam (0.4 mg/kg, reduced to 0.25 mg/kg for those already on regular anti-convulsants)

Management

Different treatment guidelines and regimen

Each department should have clear management plan

Prompt administration of effective drug in adequate doses

Monitor complications & side effects

Investigations for underlying causes eg brain malformation, infections, poisoning, overdose etc

First line treatment

0 min

0-5 min

- Supportive (semi-prone position, O2 suppl etc)
- Prepare anti-epileptic medications

6-30 min

- First line anti-epileptic medications
- Quick, convenient and effective route

5 min

Benzodiazepine (repeat bolus after 5-10min if necessary)

Diazepam

- Rectal route convenient and effective
- IM NOT effective
- Intravenous – painful

Lorazepam

- Slightly more effective compare with diazepam

Midazolam

- Buccal / intranasal admin as effective as rectal

20 min

Second line treatment if NOT responsive

20 min

Intravascular access necessary

Phenytoin (first choice)

Loading 15-20 mg/kg iv, infusion rate at < 1mg/kg/min
then 5-8mg/kg/day
side effects: strong alkaline, hypotension, arrhythmia

Phenobarbital

60 min

Loading 15-20 mg/kg iv
then 3-5mg/kg/day
side effects: respiratory suppression, hypotension

Midazolam (Continuous infusion)

Loading dose of 0.2 mg/kg as slow IV bolus, followed by infusion of up to 30 microgram/kg/min

Levetiracetam (Keppra)

getting more popular, little side effect,
less CNS depression and haemodynamic disturbance.

Refractory Status

Thiopentone or pentobarbitone infusion
Propofol / other anaesthetic agents
Sodium Valproate

Other Emergencies

Status asthmatics

Acute poisoning

Drug overdose

Cardiovascular emergency

Anaphylaxis

Surgical Considerations

Cause of sudden deterioration

- head trauma
- surgical abdomen eg. internal haemorrhage, intussusceptions, malrotation of gut.

Non-accidental injuries

Psychological considerations

- Hospitalisation especially ICU admission
- Stressful not just to patient but also immediate family members
- Support should extend to parents and family members

Summary

Close monitoring and better understanding the physiological difference between adult and children is important in early detection of acute deterioration

Master the technique paediatric CPR and know when to seek help

Understand the basic management principles on paediatric emergencies

Thank you